

GNN Semantic Segmentation of Neutrino Interactions in DUNE's TMS

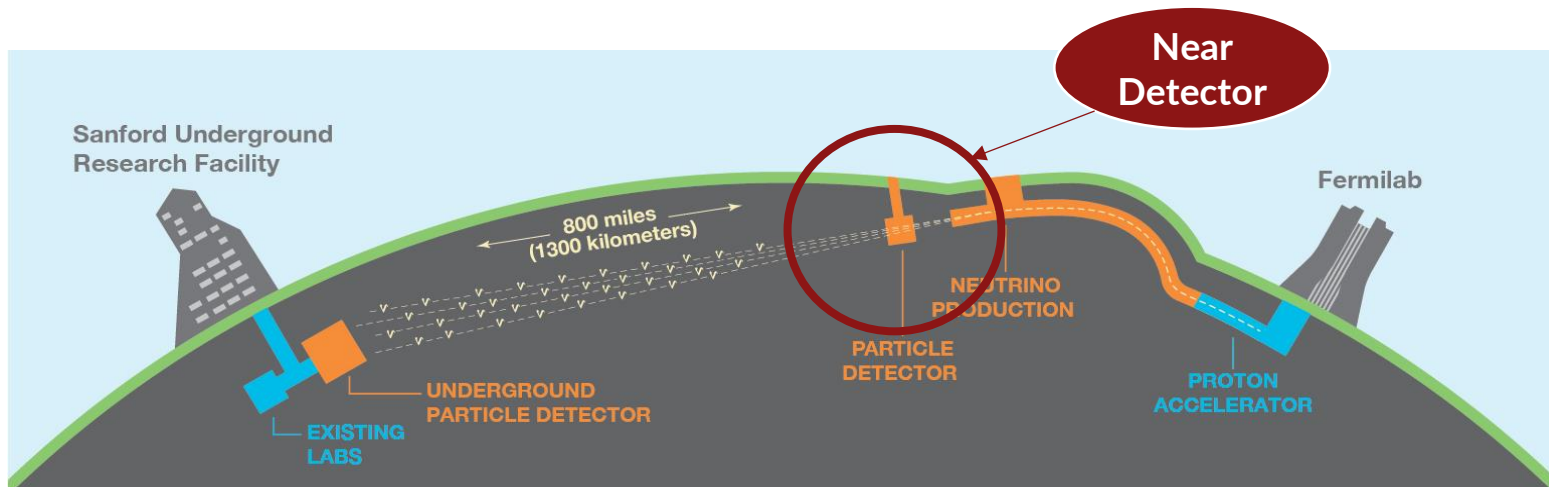
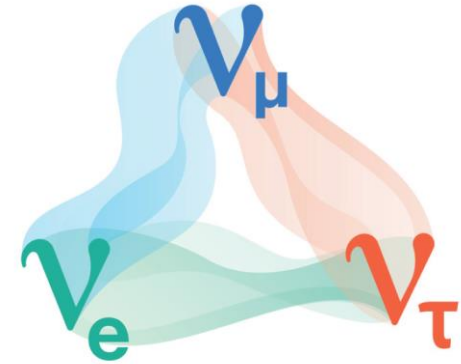
SESAPS 2025, James Madison University

Kieran Wall, for the DUNE Collaboration

10/24/2025

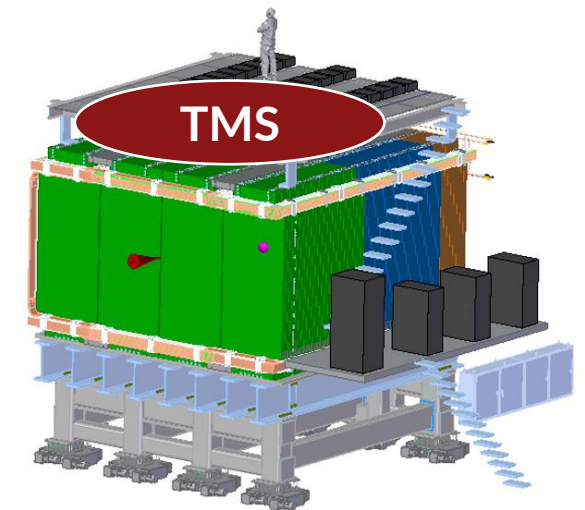
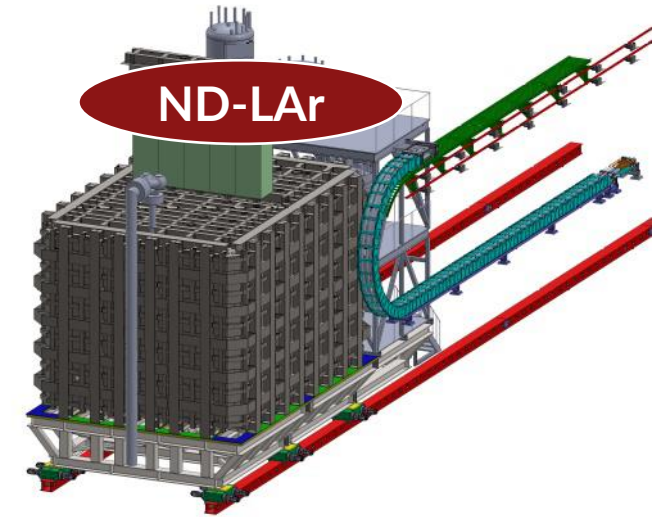
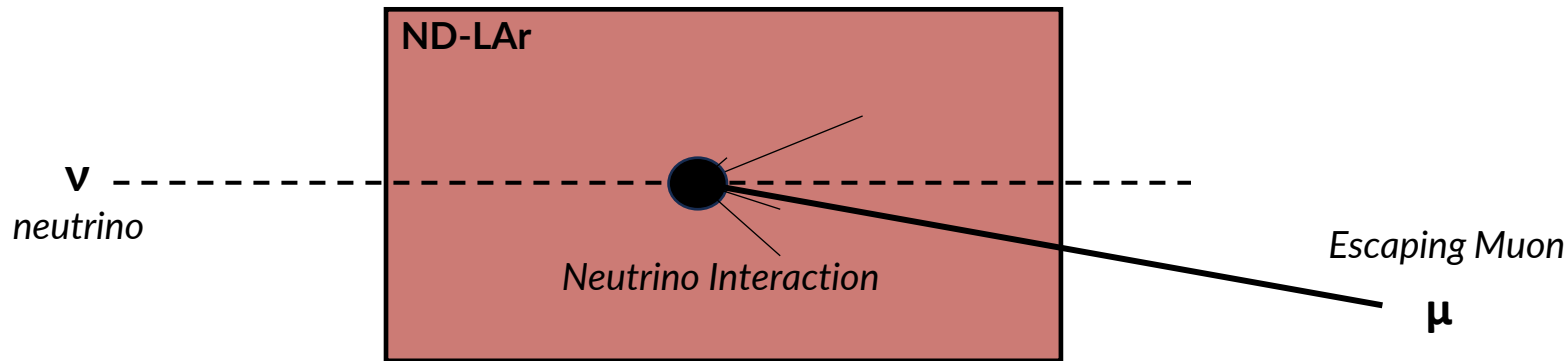
The Deep Underground Neutrino Experiment

- A next generation long-baseline neutrino oscillation experiment
 - Neutrino beam from Fermilab, **800 miles** to South Dakota
 - Particle detectors at the **Near** and **Far** ends will carefully observe and measure neutrino interactions
 - Will measure Δm^2_{32} , $\sin^2\theta_{32}$, and δ_{CP} to world leading precision



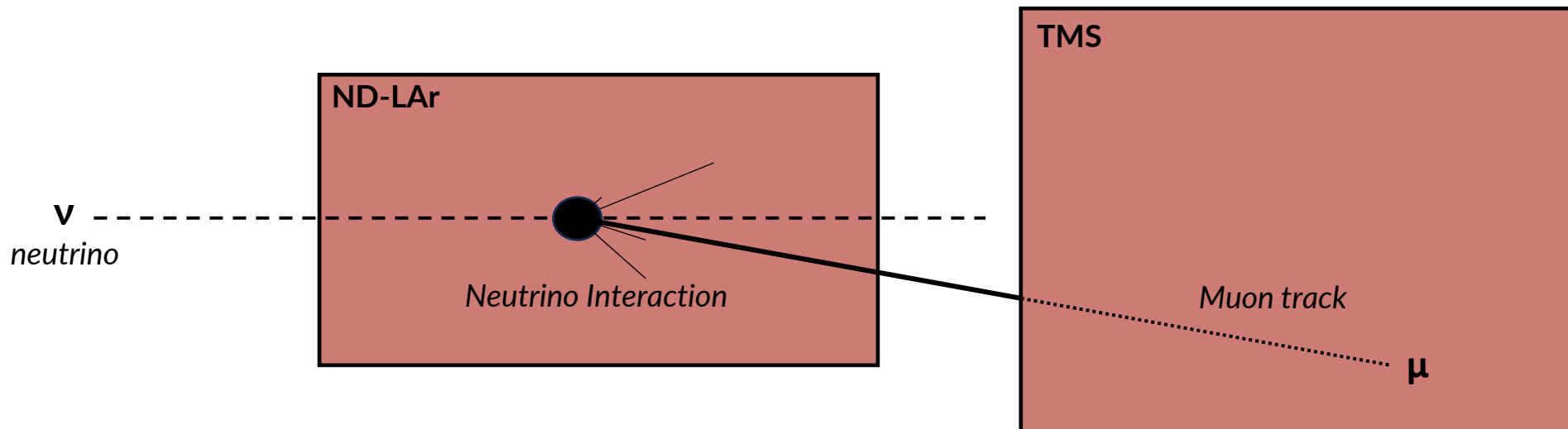
DUNE's Near Detector

- Three detectors that work together to observe neutrino interactions.
 - ND-LAr → Liquid Argon Time Projection Chamber
 - TMS (Muon Spectrometer) → Scintillator and Steel Array
 - SAND → Magnetized calorimeter for beam monitoring
- Need to measure the **energy of particles** emerging from interactions!
- High energy **muons** can escape ND-LAr, if it was the only detector, could not accomplish the full mission of the Near Detector!



The role of DUNE's TMS

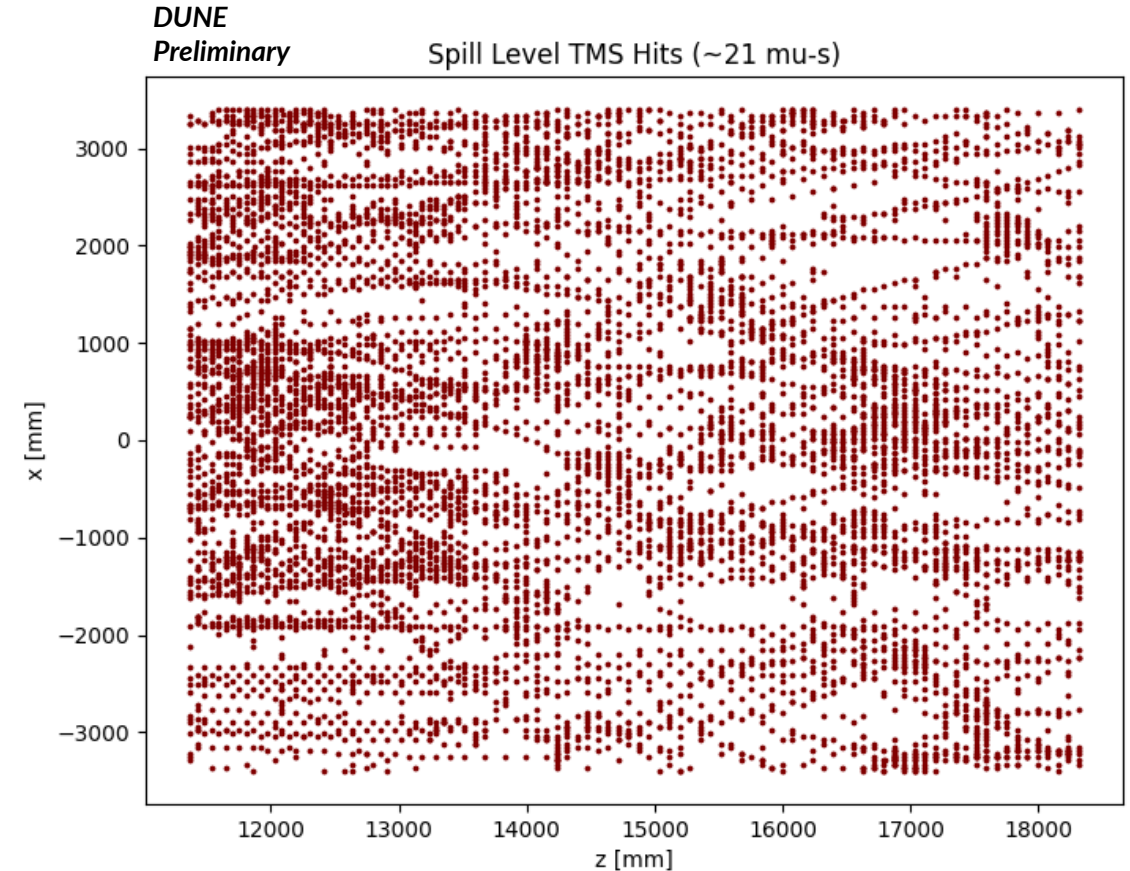
- Downstream of ND-LAr, designed to detect fleeing muons
- Muons leave tracks in the TMS scintillator bars, and lose energy as they pass through steel absorbing layers
 - Can measure energy by track length, reconstruct neutrino interaction energy
- To study neutrino oscillations, we **MUST** be able to identify muon track instances in the TMS!!



The Challenge...

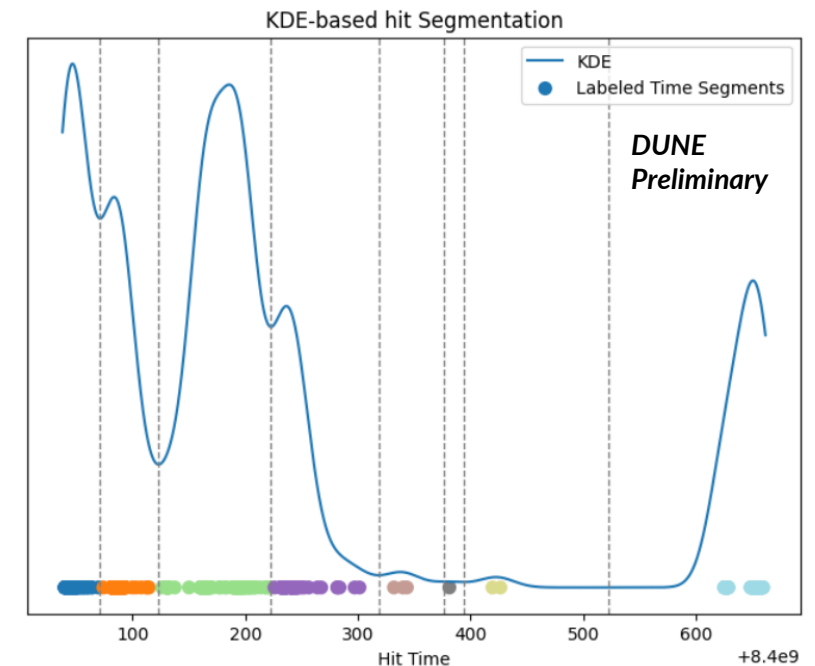
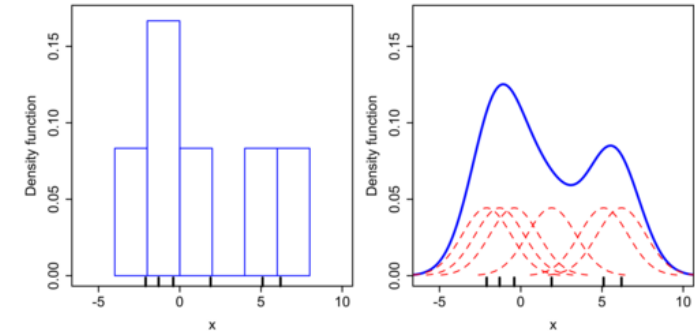
- Detailed MC simulations of LBNF beam spills model interactions in ND-LAr, the TMS, and even cave walls!
- One 'beam spill' (21 micro-seconds) worth of activity in the TMS
- There's a ton going on!!

***My Goal: Identify individual muon tracks,
with a brand-new reconstruction framework,
can ML methods deliver comparable results
to traditional reconstruction?***



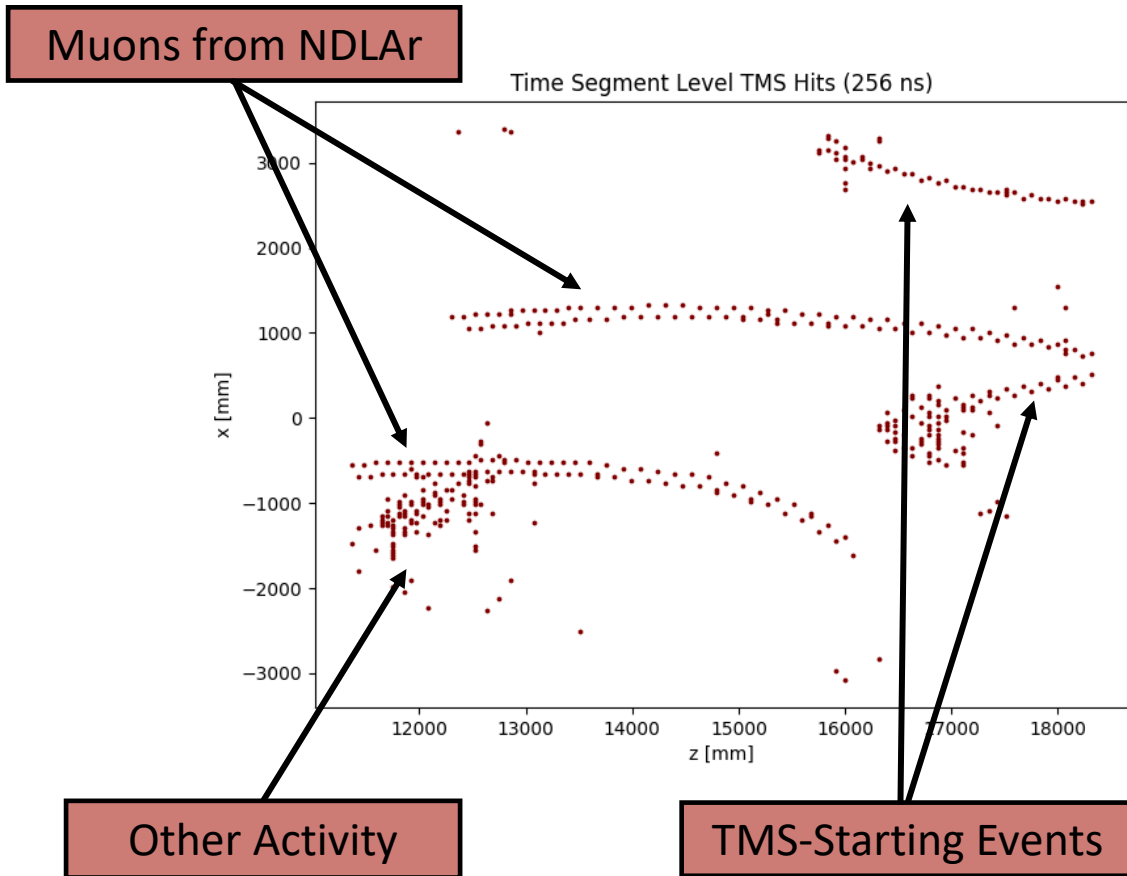
Time Segmentation – A First Step

- How do we begin the task of picking out muon tracks??
 - Employ the TMS's superpower →
~nanosecond timing resolution
- Can break our 21 micro-second spill into much more digestible chunks by grouping hits that occur close in time.
- Simple algorithm – **Kernel Density Estimation**
 - Split hits into time segments at local minima of the KDE.

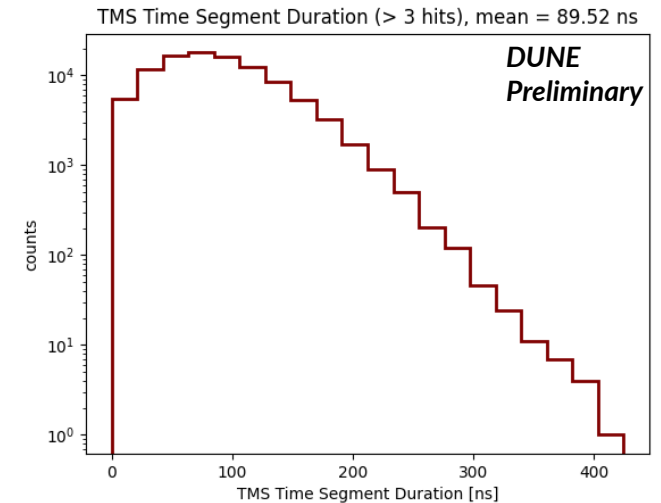


Time Segmentation

- What do our time segments look like with the KDE? – picked one with > average occupancy



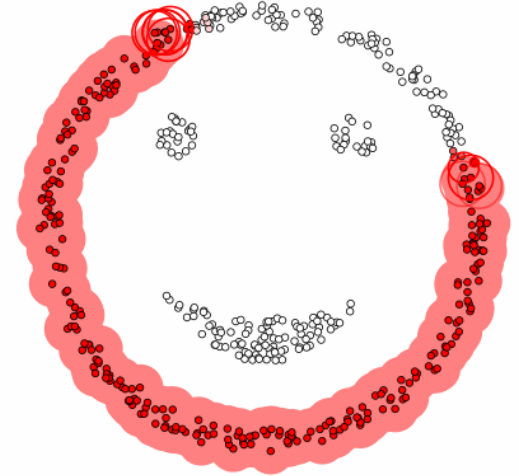
Typically finding segments of about 90ns in duration



Preliminary - Segments are 95% energy efficient, with an average of 3.75 interactions per segment

DBSCAN Clustering

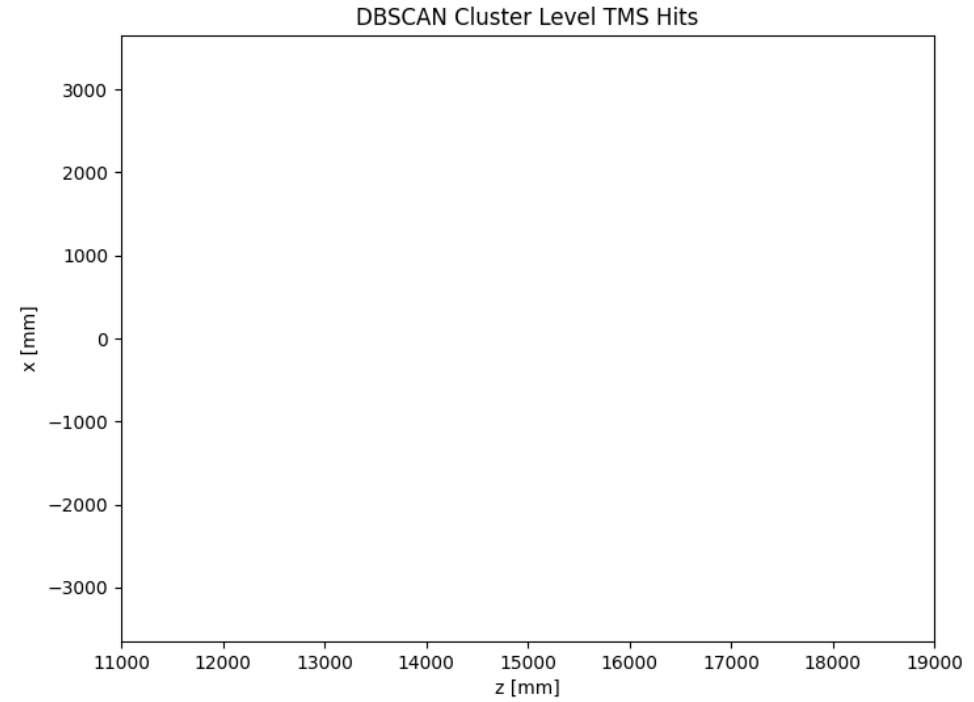
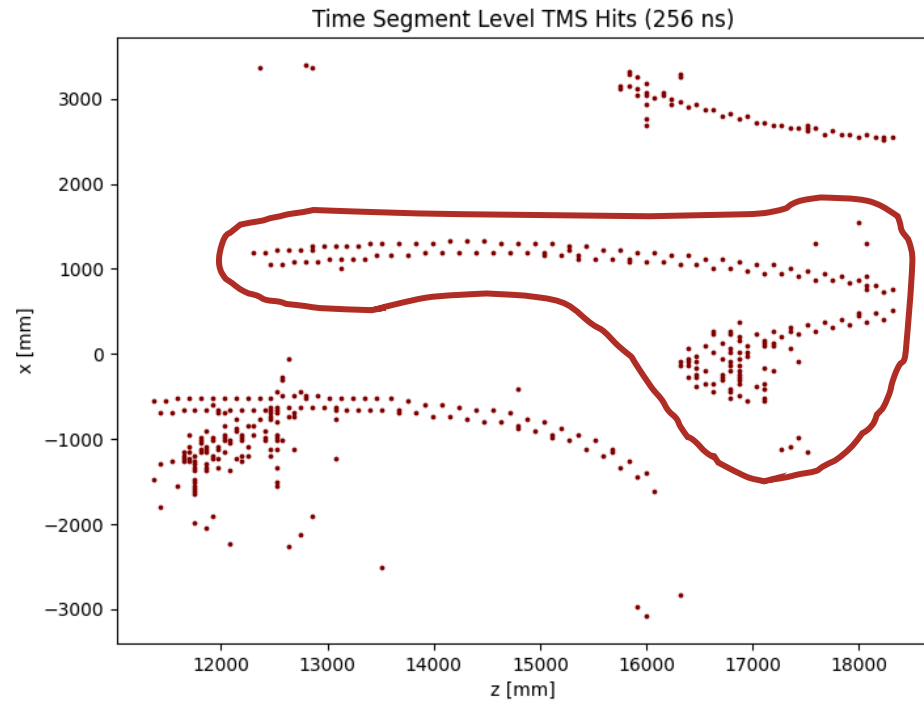
- Time segments are improving in efficiency and occupancy, but can further isolate physics instances within segments via **spatial clustering**
- **DBSCAN** (Density Based Spatial Cluster of Application with Noise) can search for non-elliptical clusters in our 2D hits.



DBSCAN in action

DBSCAN Clustering

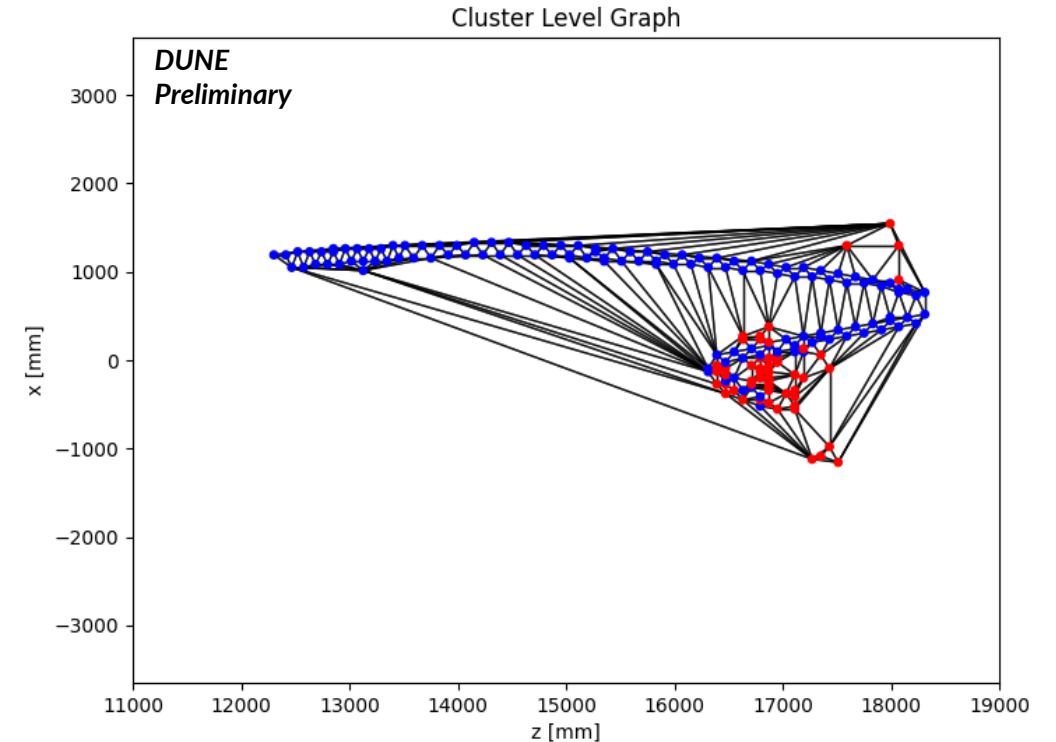
- DBSCAN breaks our time segments into more digestible chunks with lower occupancy, here's one of the 4 formed from this segment



Preliminary - Clusters are 93% energy efficient, with an average of 1.98 interactions per cluster

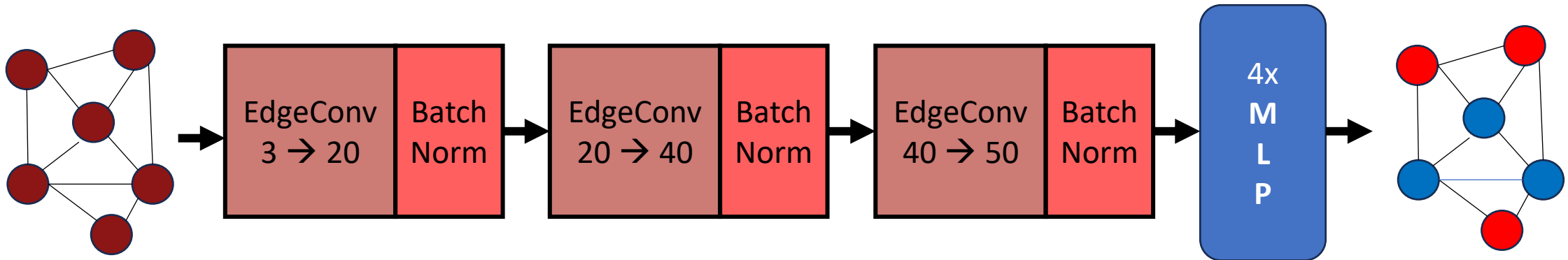
Tracks with GNN Semantic Segmentation - Graphs

- Want to extract all hits that are part of tracks:
 - Semantic segmentation → labeling hits by what macroscopic structure they belong to.
 - **Common task in computer vision**
- First step in ND-LAr's reconstruction (SPINE)
 - <https://github.com/DeepLearnPhysics/spine>
- Typically use a CNN → I use GNNs due to TMS' non-uniformities and odd geometry
 - Graph Convolutions preserve strengths of CNNs.
 - Can efficiently connect hits to form a graph using **Delaunay Triangulation**
- Assign truth labels of **track** vs **other**



Tracks with GNN Semantic Segmentation - Model

- Input – graph of nodes with feature vector (z,x,PE)
- Convolutional layers → 3 layers of **Edge Convolution Blocks**, imbed from 3 to 50 dim
 - Message passing extends to 3rd order neighborhood
 - Batch normalization for regularization
- 4-layer MLP to map convolved nodes down to 2 dim, which correspond to labels track or other

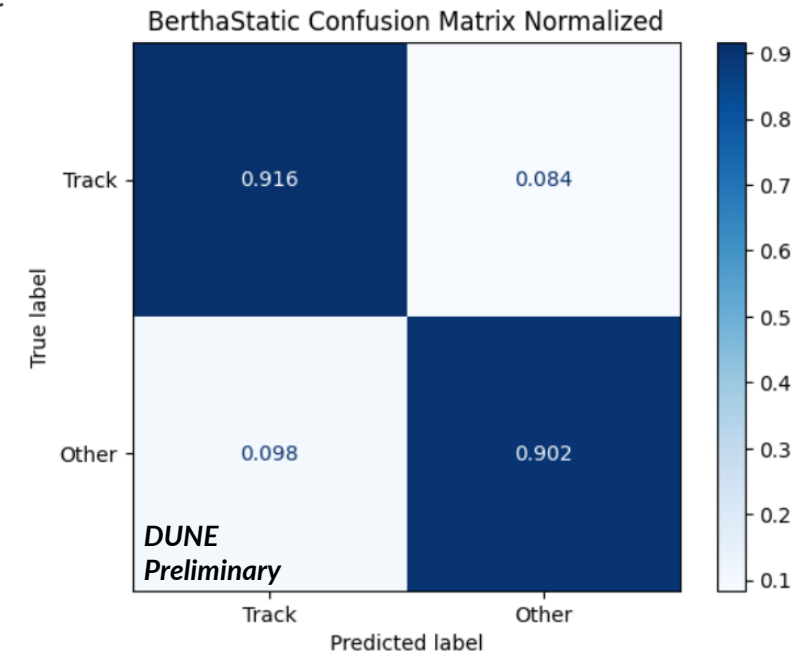
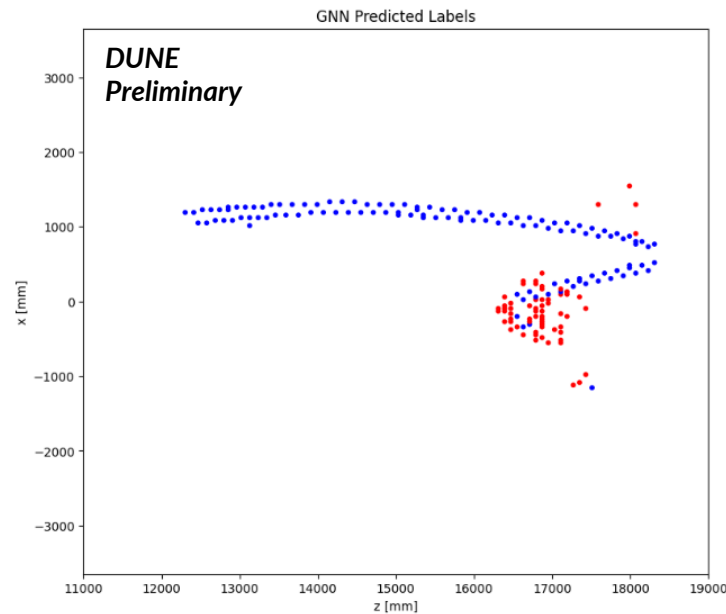
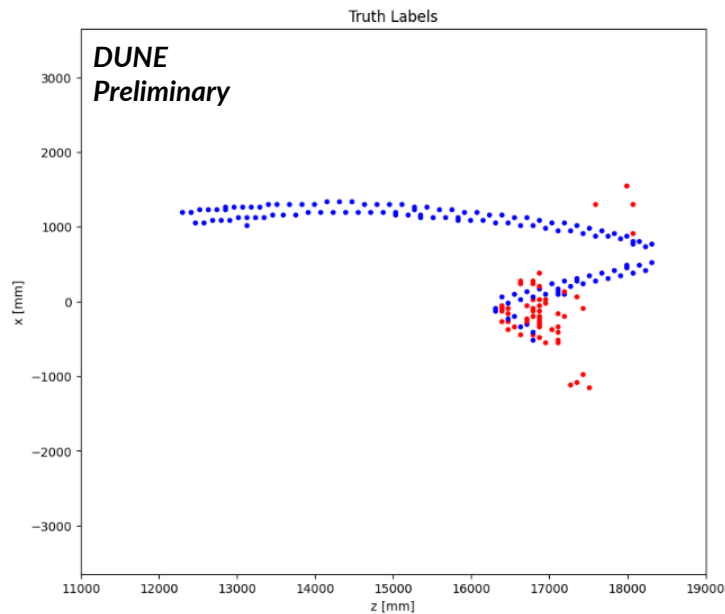


A simple, lightweight model

Tracks with GNN Semantic Segmentation - Performance

- Trained on >200k labeled graphs, tested on 12k
 - Fairly strong performance at **track** vs **other** task!
 - The example we have been following →
 - Not perfect, loses the smaller track in the TMS starting event

91% true-positive rate for track instances, relatively low false positive rate



Tracks with GNN Semantic Segmentation – What's Next?

- This work could aid the TMS's muon tracking task, facilitating DUNE's oscillation analysis!
- There's still room for optimization and experimentation at the GNN architecture and training level for this **track** vs **other** labeling
- Extracting tracks → need to group track labeled hits together into instances
- 3D reconstruction → *In progress!*
 - Ideal world, this would go *before* a labeling GNN, so could take advantage of 3D features, almost certainly will be necessary for a more complex labeling
- With more work, could serve as an alternative reconstruction pipeline to existing framework

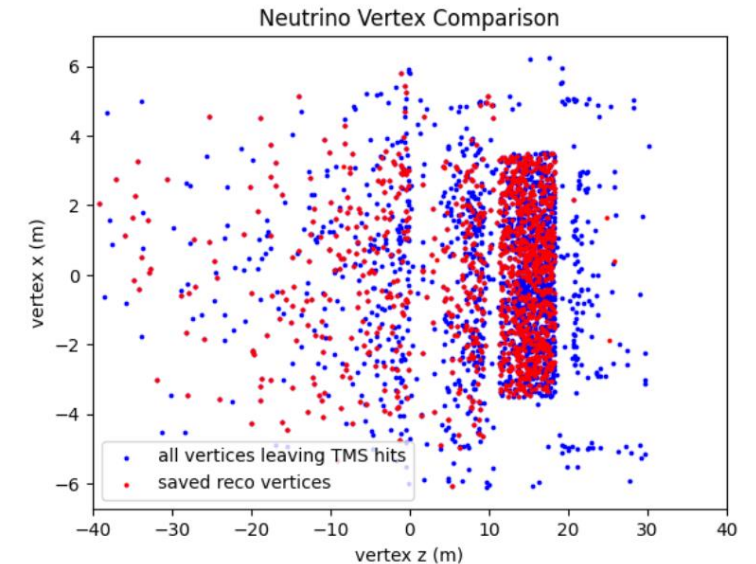
Thanks!



This work was made possible through the Department of Energy Office of Science's SULI Program

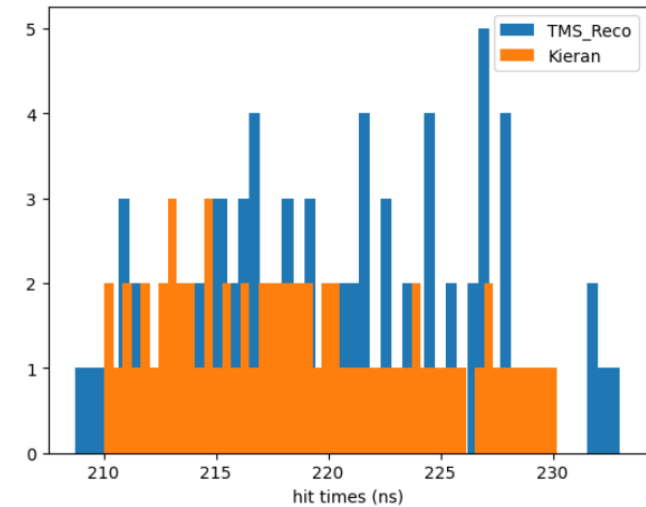
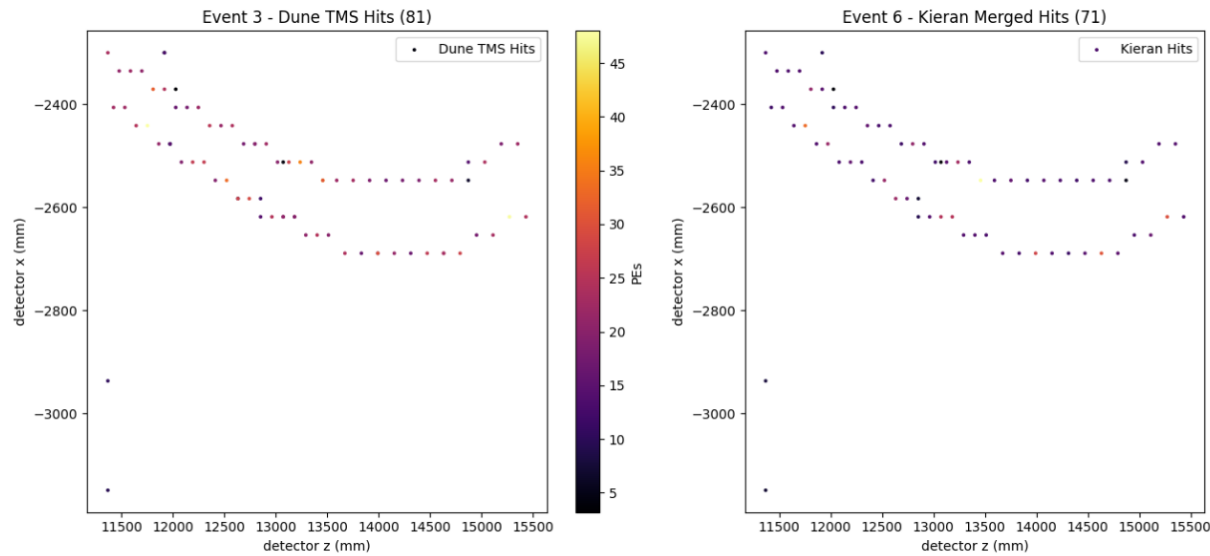
Backup – Neutrino Vertex Weirdness in TMS Reco

- Want to be able to trace every TMS hit back to the neutrino vertex that caused it
- Investigated the TrueRecoHits saved in TMS Reco tree – *NeutrinoX4 collection*
 - Example file: full spill 2459
 - 1266 neutrino vertices – matches number of ‘events’
 - Only 1145 unique vertices, not infeasible that this could be events happening at same places though.
- Perhaps more importantly **1266** ‘events’ saved in Reco, but **2625** neutrino interactions leave activity in TMS. Want to study all these interactions!

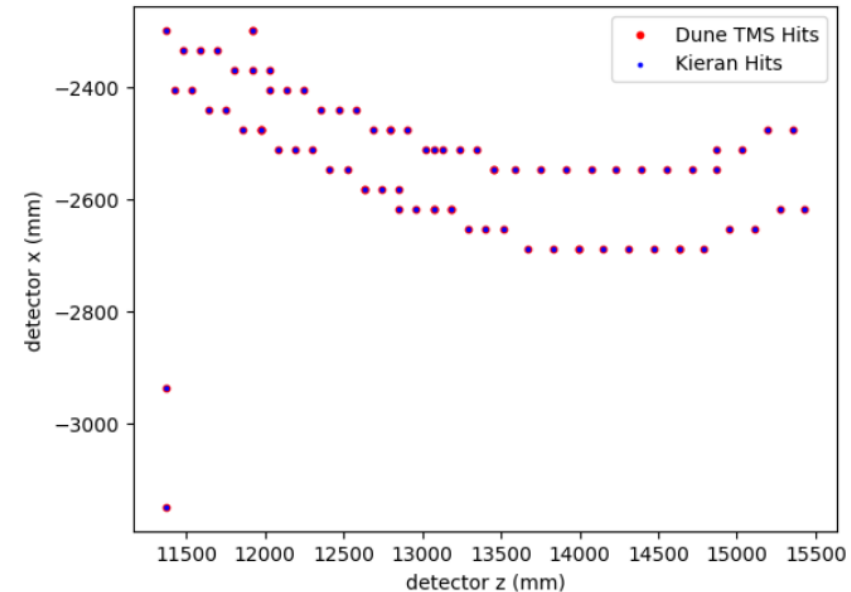


To be clear, not asserting anything is fundamentally broken in the reco, but I was left confused enough and unsure about hit mapping to vertices to build an in-house framework

Backup – Comparing dune-tms and my sim



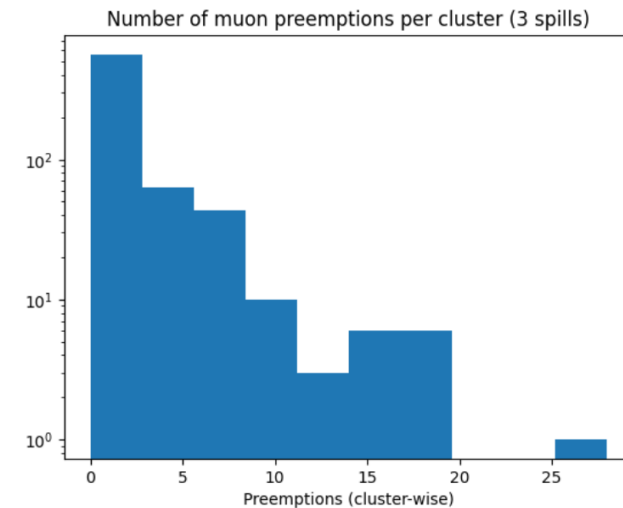
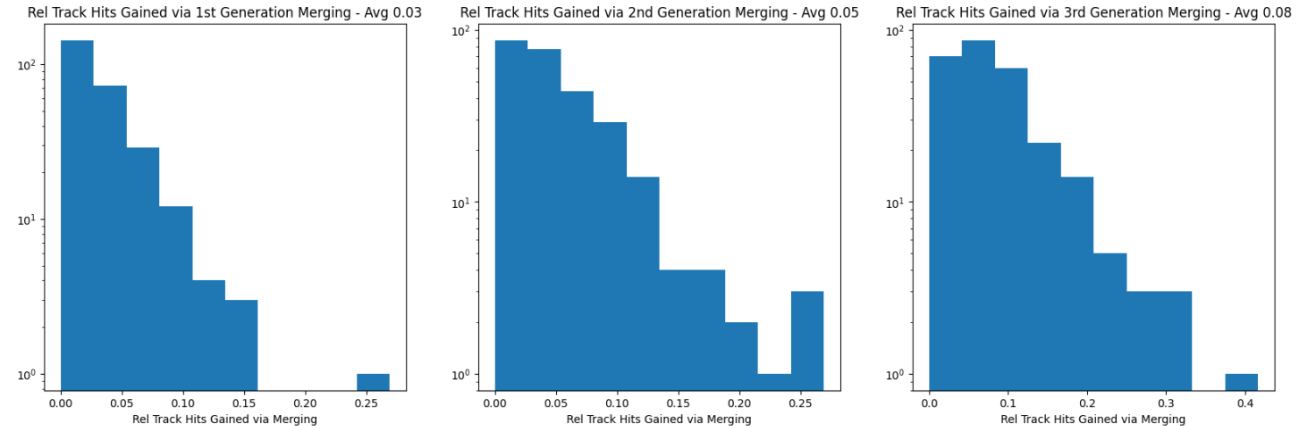
Hit merging is handled slightly differently by my code, so not shocked at a discrepancy in number, most important is that times are not completely different, no weirdness in position. Time slew likely from different RNG
-Slight difference in PE dist, likely result of merging differences!



Backup – Semantic Labeling Logic

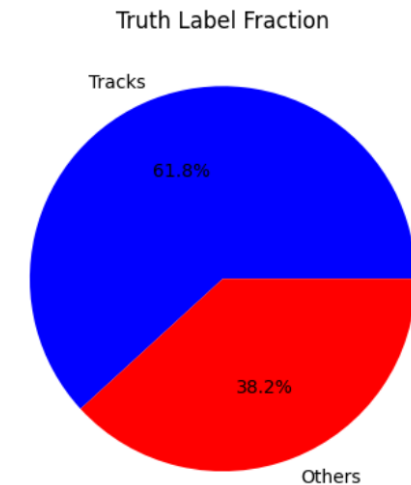
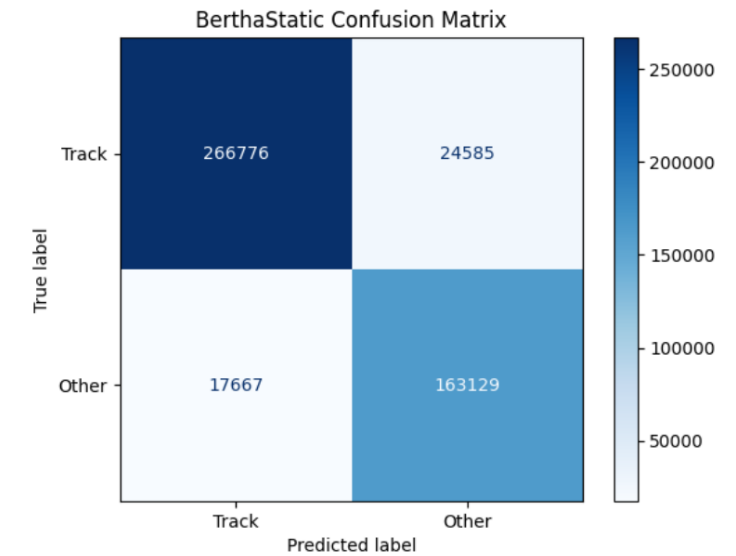
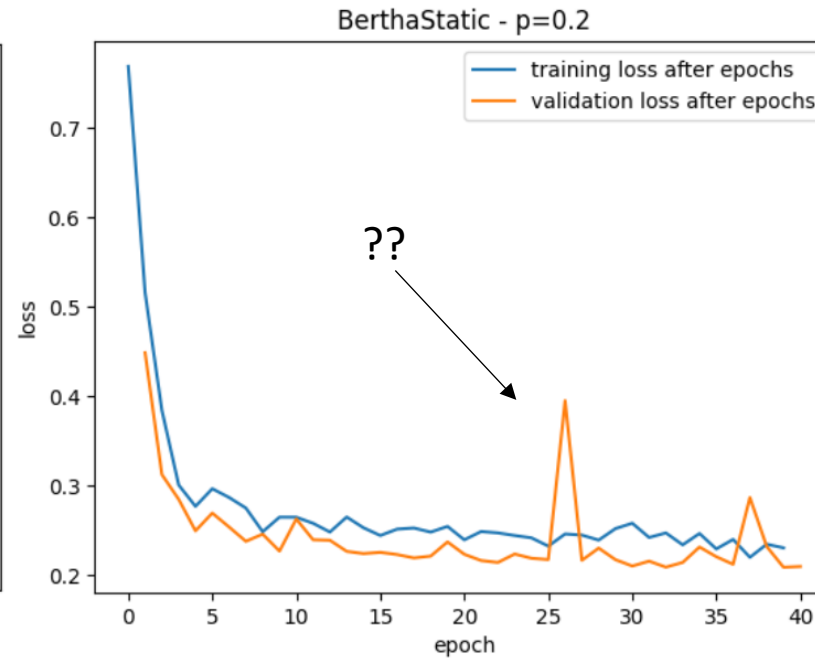
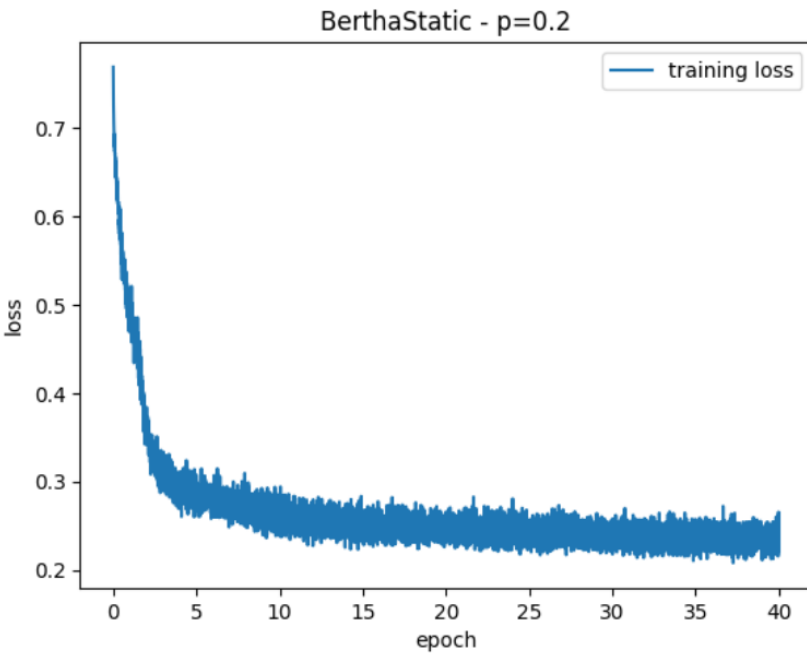
- 1) Merged hit muon preemption
- 2) Form Geant4 trackid groups
- 3) Trackid group size > 4? → **Track**, Else? → **Other**
- 4) Other Scanning:
 - Parent trackid a Track? → **Track**
 - Grandparent trackid a Track? → **Track**
 - Great-Grandparent trackid a Track? → **Track**
 - Else? → **Other**

Designed to grab extended tracks from muons as well as some mesons and hadrons. Other scanning for track relatives facilitates merging of deltas, michels, and other track caused activity.



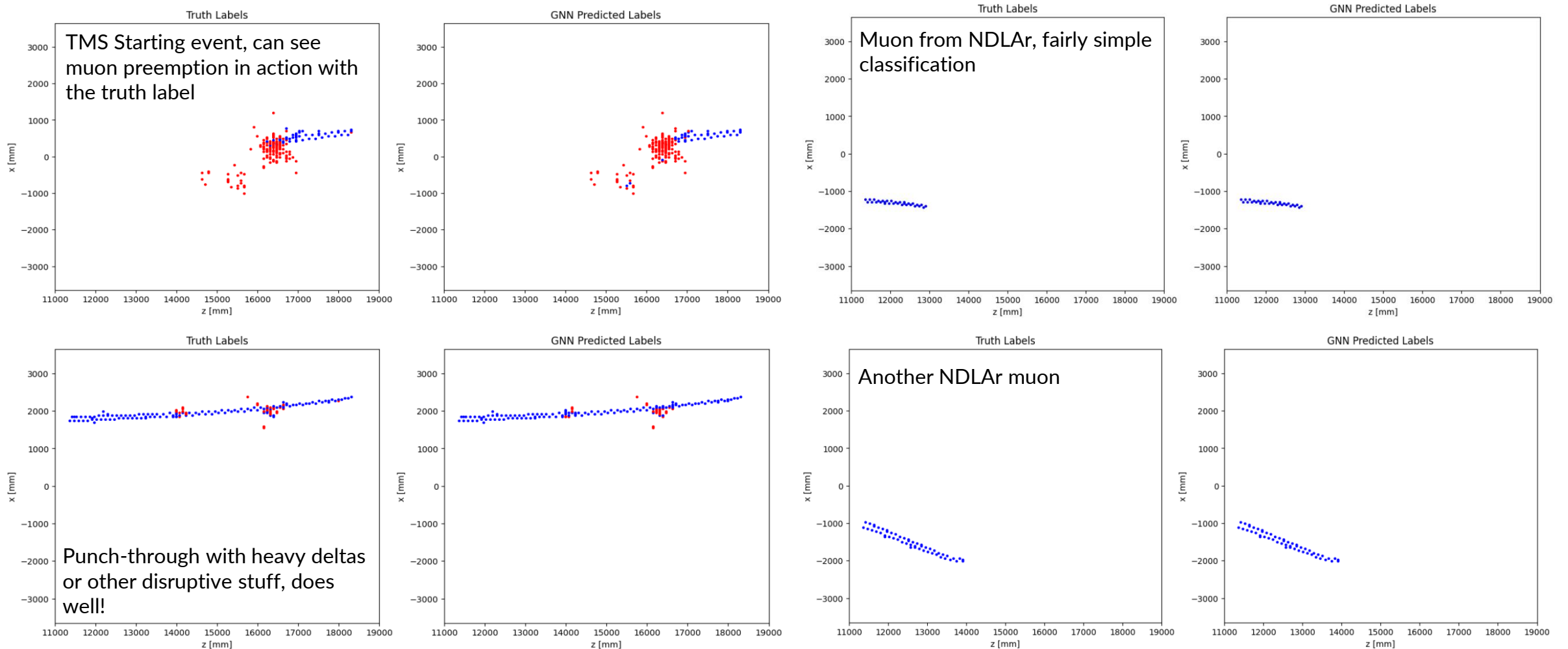
Backup – Learning Curves

- Learning curves for the model presented, as well as some related plots
- Validation loss is lower than training loss because of dropout and batch normalization



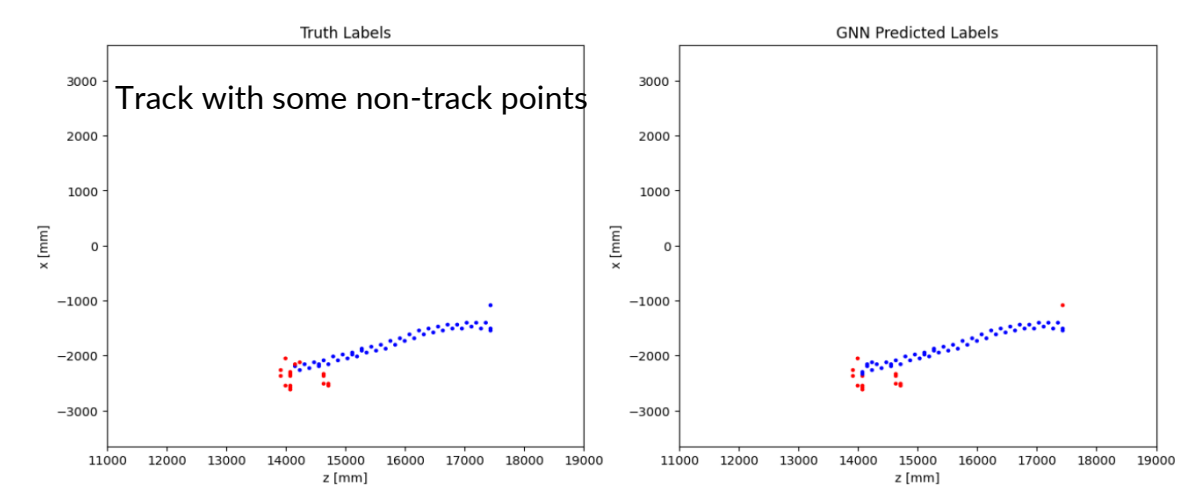
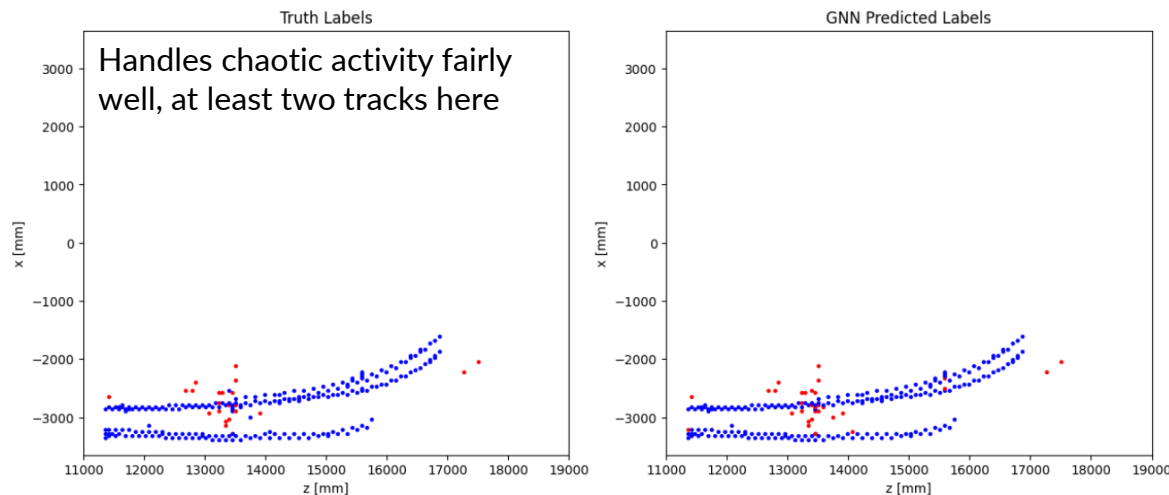
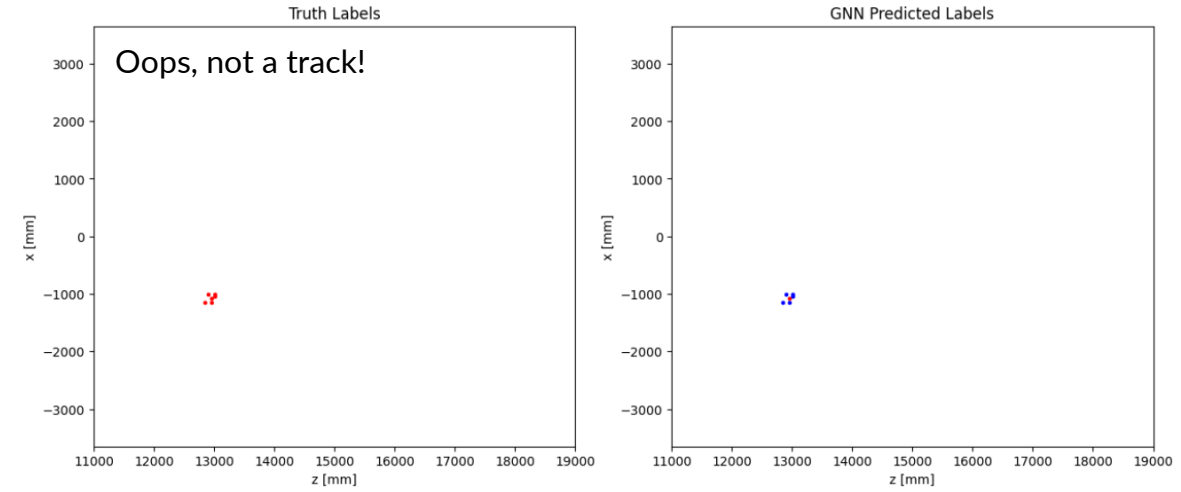
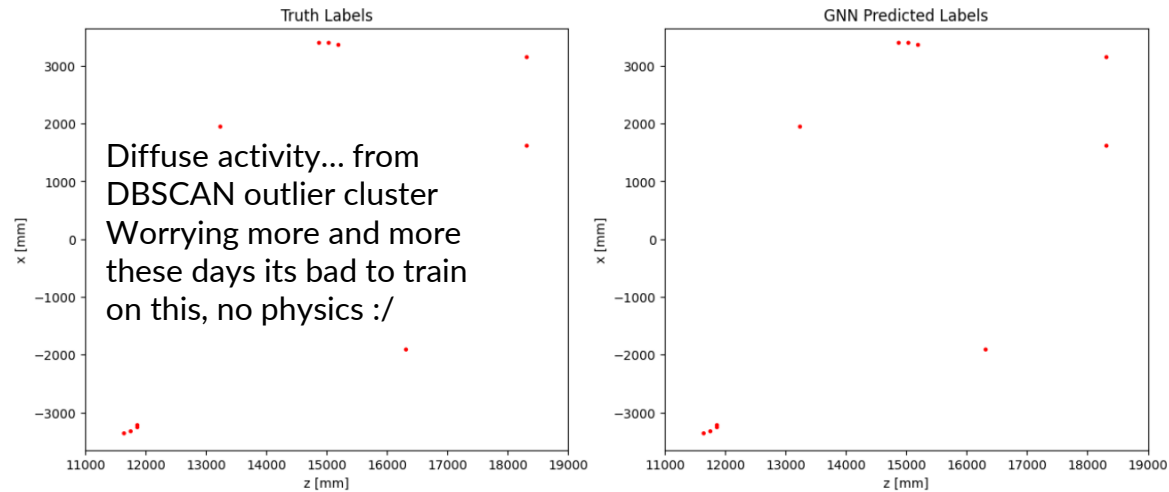
Backup – More GNN Labeling Comparisons

- Some more comparisons:



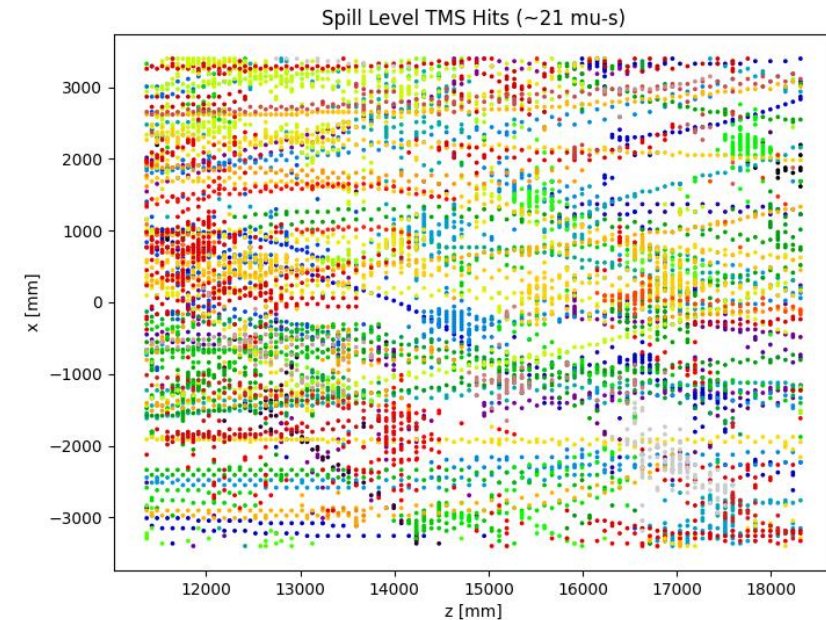
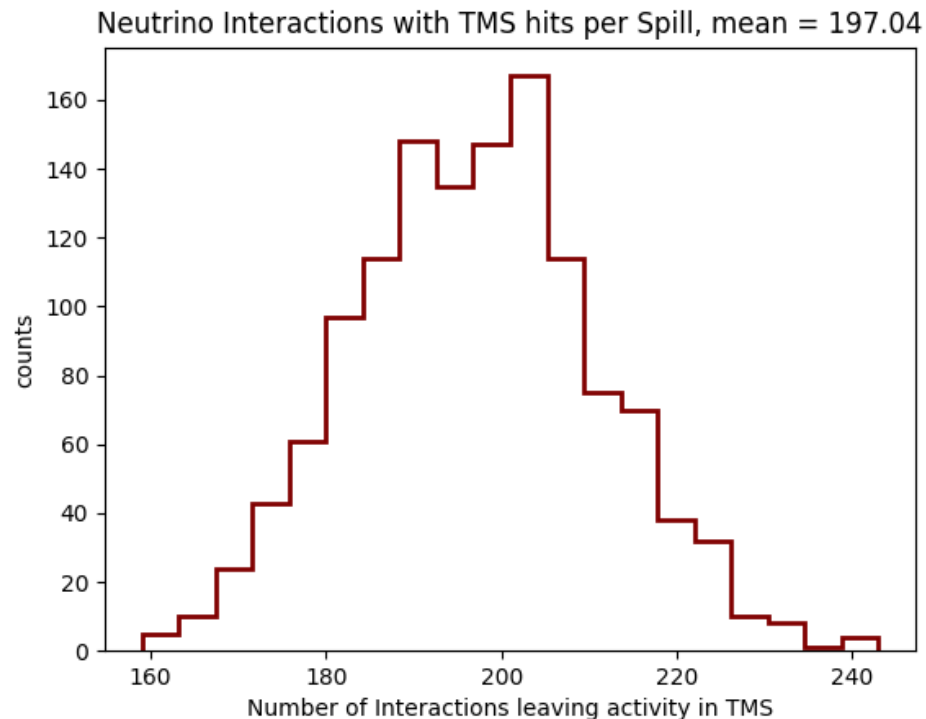
Backup – More GNN Labeling Comparisons

- Some more comparisons:



Backup - The Full-Spill Pileup Environment

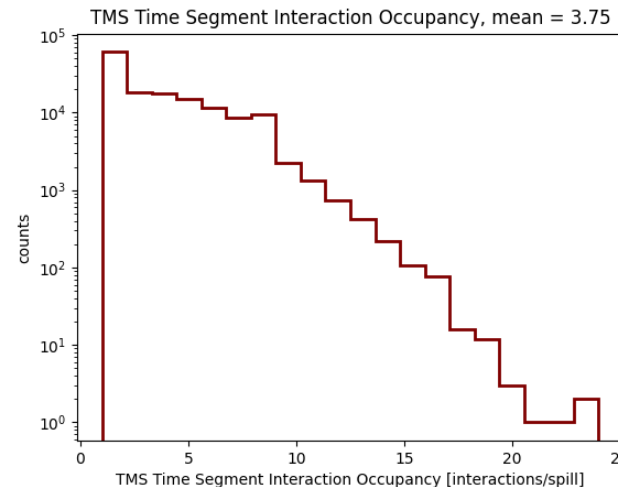
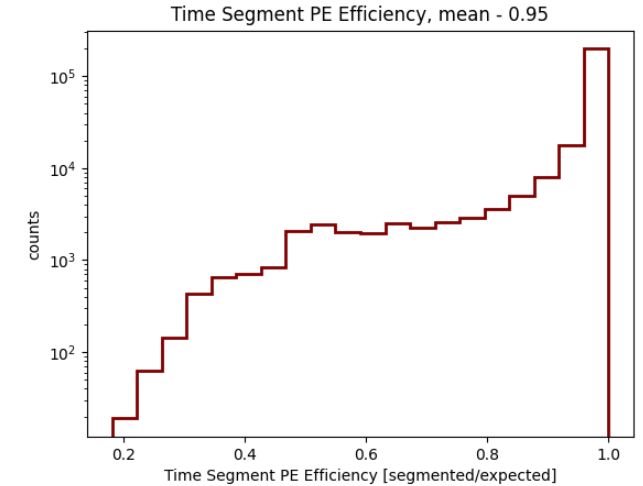
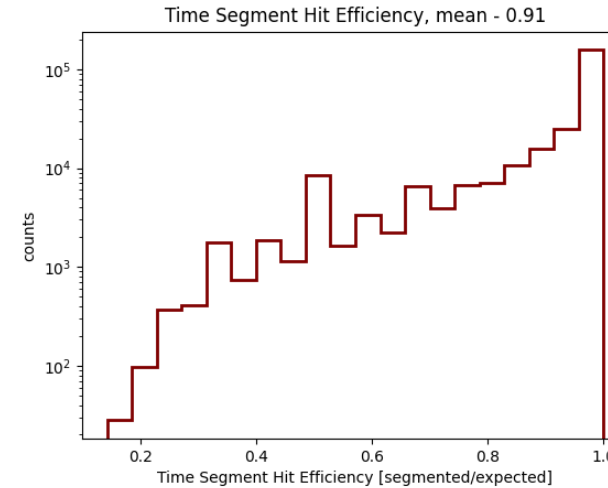
- Average of ~ 200 interactions leaving energy depositions in TMS per beam spill



- Single spill worth of activity in the TMS (~ 10 microsecond bunch leaves hits over ~ 20 microseconds)
- **Goal – extract individual muon tracks**

Backup – KDE Performance

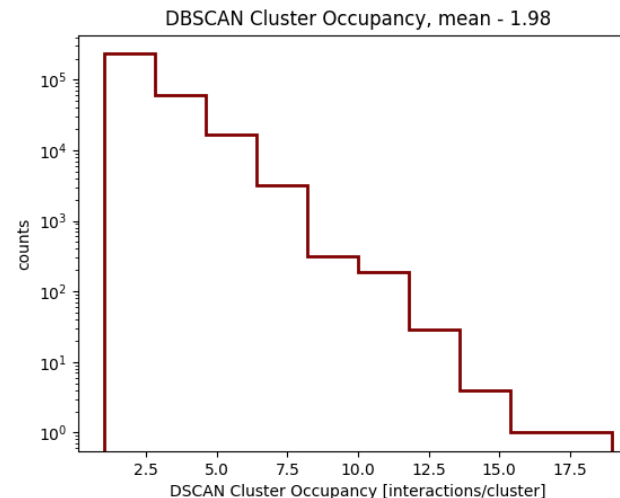
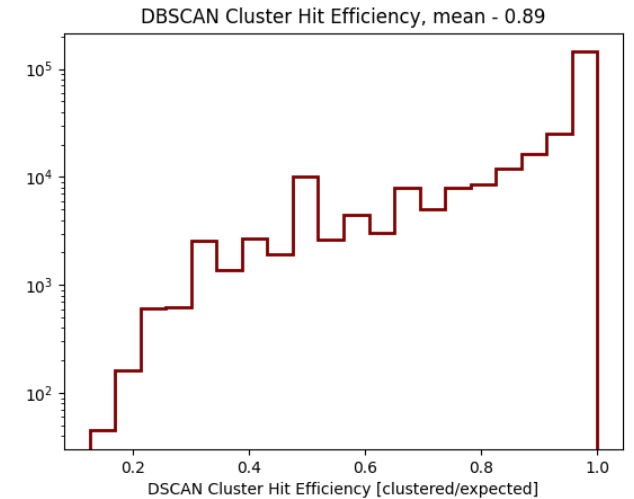
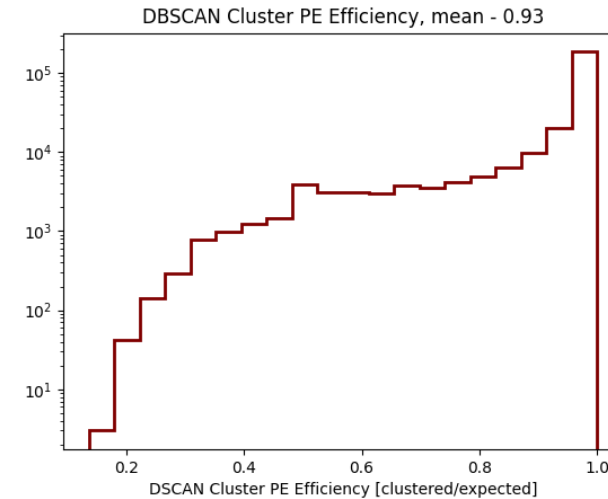
- KDE w/ $\sigma = 20$ ns, 10k mesh points applied to 1303 spills:
 - All hits segmented, with 91% hit efficiency, 95% PE efficiency, 3.75 average occupancy!
- Comparable efficiency to low threshold sliding window, comparable occupancy to high threshold sliding window



Occupancy is unique neutrino interactions producing hits / segment, efficiency is computed by interaction, hits in segment / total TMS hits caused by interaction (analogous for PE)

Backup - DBSCAN Performance

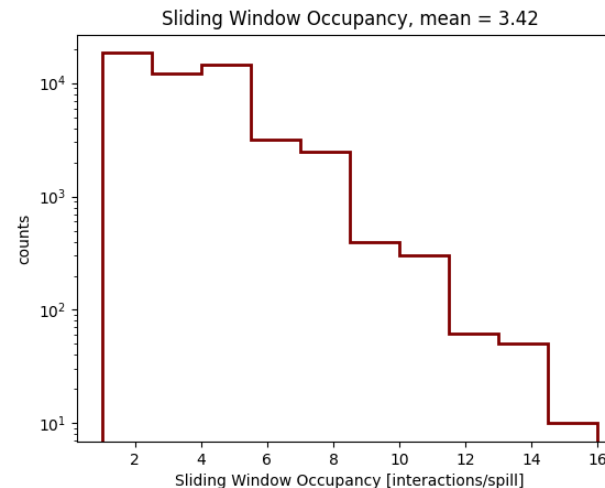
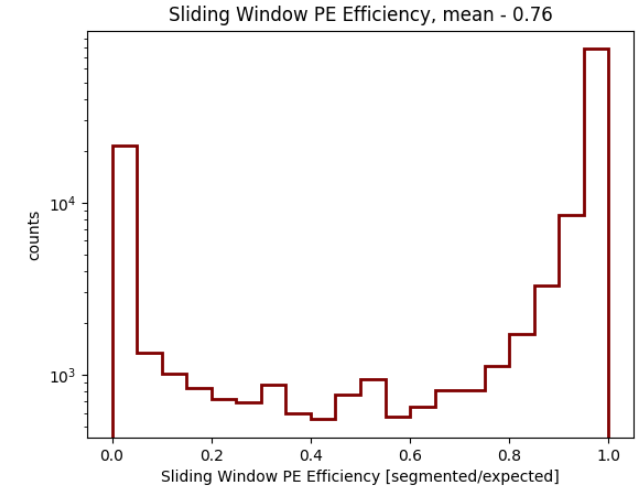
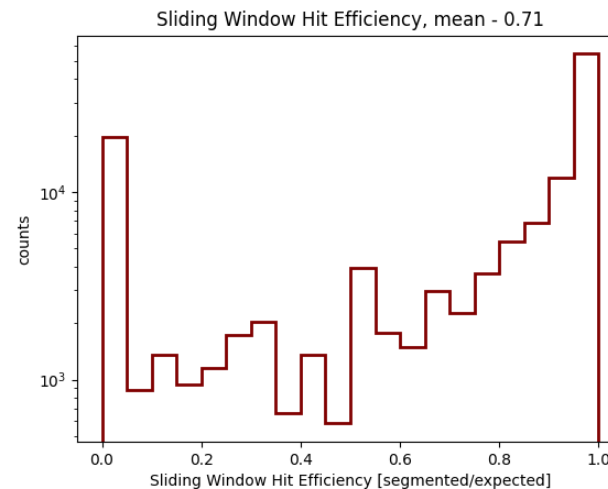
- Exchange some hit and PE efficiency for > 2 interactions per DBSCAN cluster!
- An average of ~ 240 DBSCAN clusters / beam spill
- As seen on previous slide, not perfect singular track instances, but certainly lower occupancy than time segments



Occupancy is unique neutrino interactions producing hits / segment, efficiency is computed by interaction, hits in segment / total TMS hits caused by interaction (analogous for PE)

Backup - Time Segmentation/Slicing – Sliding Window

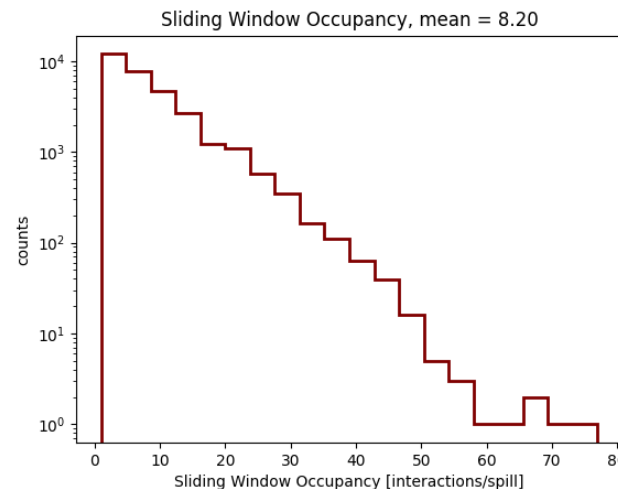
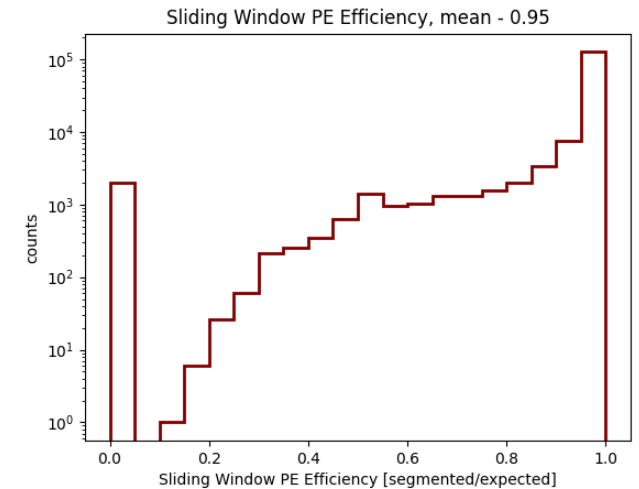
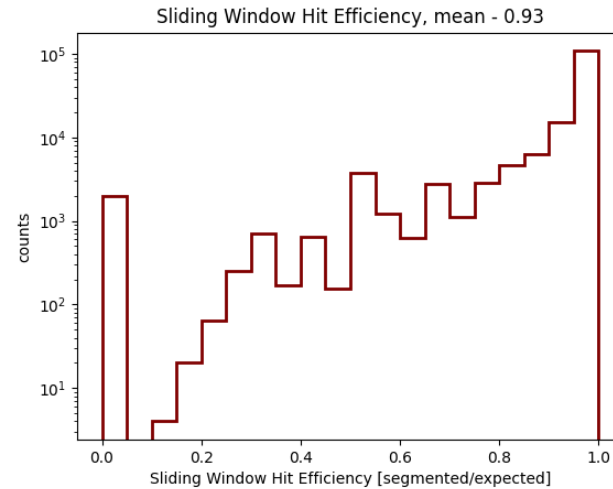
- Implemented a **sliding window algorithm** mimicking dune-tms to test performance on the full spill files.
 - 10 MeV initialization threshold (~500 PE), 5 MeV drop threshold (~250 PE)
 - Thresholds cited in the TMS TDR
- Efficiencies dragged down by interactions being missed by the slicer (the 0 bin),
 - Outlier slice is suboptimal, would be very high occupancy



Occupancy is unique neutrino interactions producing hits / segment, efficiency is computed by interaction, hits in segment / total TMS hits caused by interaction (analogous for PE)

Backup Time Segmentation/Slicing – Sliding Window

- Experimented with lowering the thresholds:
 - Initialization: 8 PE
 - Drop: 4 PE
- Better efficiencies, higher occupancy, **tradeoff for any time segmentation!**
 - *Can we develop a time segmentation method that has high efficiency, lower occupancy, and segments all hits, not just ones in a triggered window?*



Occupancy is unique neutrino interactions producing hits / segment, efficiency is computed by interaction, hits in segment / total TMS hits caused by interaction (analogous for PE)